



This Booklet contains 24 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are 720.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Facsimile signature stamp of

Centre Superintendent : _____

**ENTHUSIAST & LEADER COURSE
(PHASE : ALL)**

ALLEN MAJOR TEST

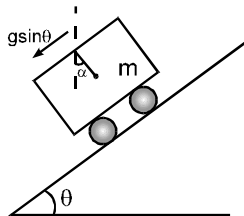
DATE : 21 - 03 - 2020

SYLLABUS – 01 & 02

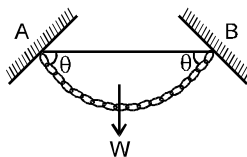
1. A particle starts from rest with uniform acceleration a . Its velocity after n seconds is v . The displacement of the body in the last two seconds is :

(1) $\frac{2v(n-1)}{n}$ (2) $\frac{v(n-1)}{n}$
(3) $\frac{v(n+1)}{n}$ (4) $\frac{2v(2n+1)}{n}$

2. A trolley is accelerating down an incline of angle θ with acceleration $g \sin \theta$. Which of the following is correct. (α is the angle made by the string with vertical).

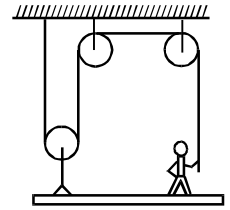


- (1) $\alpha = \theta$
(2) $\alpha = 0^\circ$
(3) Tension in the string, $T = mg$
(4) Tension in the string, $T = mg \sec \theta$
3. A flexible chain of weight W hangs between two fixed points A and B at the same level. The inclination of the chain with the horizontal at the two points of support is θ . What is the tension of the chain at the endpoint.



(1) $\frac{W}{2} \operatorname{cosec} \theta$
(2) $\frac{W}{2} \sec \theta$
(3) $W \cos \theta$
(4) $\frac{W}{3} \sin \theta$

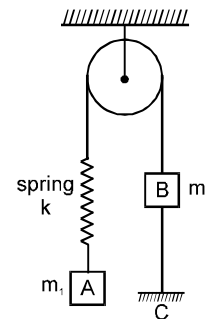
4. A 50 kg person stands on a 25 kg platform. He pulls on the rope which is attached to the platform via the frictionless pulleys as shown in the fig.



The platform moves upwards at a steady rate if the force with which the person pulls the rope is

- (1) 500 N (2) 250 N
(3) 25 N (4) 50 N

5. In the system shown in the figure $m_1 > m_2$. System is held at rest by thread BC. Just after the thread BC is burnt :

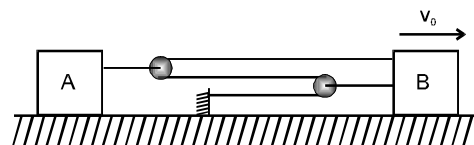


- (1) acceleration of m_2 will be downwards
(2) magnitude of acceleration of both

blocks will be equal to $\left(\frac{m_1 - m_2}{m_1 + m_2} \right) g$

- (3) acceleration of m_1 will be equal to zero
(4) magnitude of acceleration of two blocks will be non-zero and unequal.

6. Block B moves to the right with a constant velocity v_0 . The velocity of body A relative to B is :

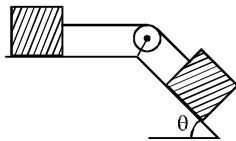


- (1) $\frac{v_0}{2}$, towards left
(2) $\frac{v_0}{2}$, towards right
(3) $\frac{3v_0}{2}$, towards left
(4) $\frac{3v_0}{2}$, towards right

7. Five persons A, B, C, D & E are pulling a cart of mass 100 kg on a smooth surface and cart is moving with acceleration 3 m/s^2 in east direction. When person 'A' stops pulling, it moves with acceleration 1 m/s^2 in the west direction. When person 'B' stops pulling, it moves with acceleration 24 m/s^2 in the north direction. The magnitude of acceleration of the cart when only A & B pull the cart keeping their directions same as the old directions, is :

- (1) 26 m/s^2 (2) $3\sqrt{71} \text{ m/s}^2$
(3) 25 m/s^2 (4) 30 m/s^2

8. Two bodies of identical mass are tied by an ideal string which passes over an ideal pulley. The co-efficient of friction between the bodies and the plane is μ . The minimum value of θ for which the system starts moving is:



- (1) $\cos^{-1}\left(\frac{\mu^2 - 1}{\mu^2 + 1}\right)$
(2) $\cos^{-1}\left(\frac{\mu^2 + 1}{1 - \mu^2}\right)$
(3) $\cos^{-1}\left(\frac{2\mu}{1 + \mu^2}\right)$
(4) $\theta = \cos^{-1}\left(\frac{1 - \mu^2}{1 + \mu^2}\right)$

9. When a rigid body of mass M slides down an inclined plane of inclination θ , having coefficient of friction μ through a distance s , the work done against friction is :

- (1) $\mu (Mg \cos \theta) s$
(2) $\mu (Mg \sin \theta) s$
(3) $Mg (\mu \cos \theta - \sin \theta) s$
(4) None of the above

10. Two springs have their force constant as k_1 and k_2 ($k_1 > k_2$). When they are stretched by the same force

- (1) No work is done by this force in case of both the springs
(2) Equal work is done by this force in case of both the springs
(3) More work is done by this force in case of second spring
(4) More work is done by this force in case of first spring

11. A heavy stone is thrown from a cliff of height h with a speed v . The stone will hit the ground with maximum speed if it is thrown

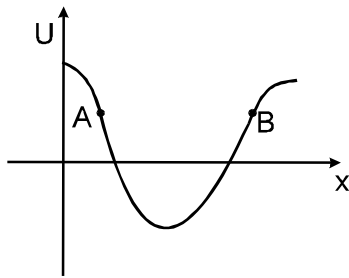
- (1) vertically downward
(2) vertically upward
(3) horizontally
(4) the speed does not depend on the initial direction.

12. Figure shows a particle sliding on a frictionless track which terminates in a straight horizontal section. If the particle starts slipping from the point A, how far away from the track will the particle hit the ground ?

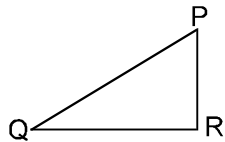


- (1) At a horizontal distance of 1 m from the end of the track.
(2) At a horizontal distance of 2 m from the end of the track.
(3) At a horizontal distance of 3 m from the end of the track.
(4) Insufficient information
13. An electric motor creates a tension of 4500 N in hoisting cable and reels it at the rate of 2 m/s. What is the power of electric motor ?
- (1) 9 W
(2) 9 KW
(3) 225 W
(4) 9000 H.P.

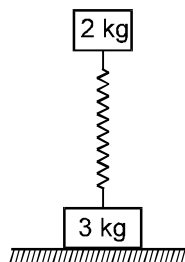
14. Potential energy v/s displacement curve for one dimensional conservative field is shown. Force at A and B is respectively.



- (1) Positive, Positive
(2) Positive, Negative
(3) Negative, Positive
(4) Negative, Negative
15. For the path PQR in a conservative force field (fig.), the amounts work done in carrying a body from P to Q & from Q to R are 5 J & 2 J respectively. The work done in carrying the body from P to R will be -

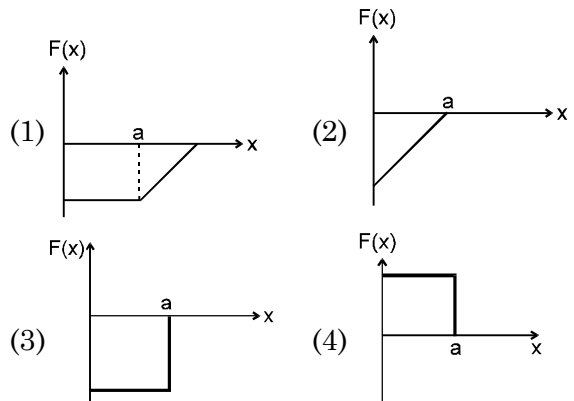
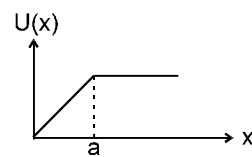


- (1) 7 J
(2) 3 J
(3) $\sqrt{21}$ J
(4) zero
16. The ends of a spring are attached to blocks of masses 3 kg and 2 kg. The 3 kg block rests on a horizontal surface and the 2 kg block which is vertically above it is in equilibrium producing a compression of 1 cm of the spring. The value of the length to which the 2 kg mass must be compressed, so that when it is released, the 3 kg block may be lifted off the ground is :

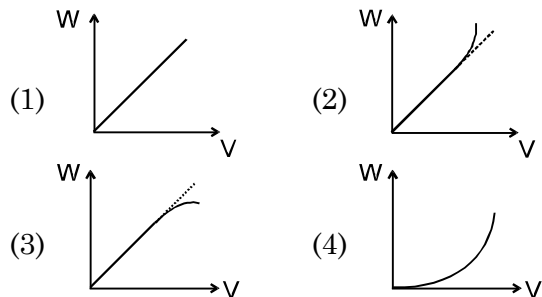


- (1) 5 cm
(2) 2.5 cm
(3) 8 cm
(4) 10 cm

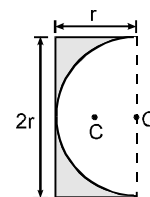
17. The potential energy of a system is represented in the first figure, the force acting on the system will be represented by



18. A particle, initially at rest on a frictionless horizontal surface, is acted upon by a horizontal force which is constant in size and direction. A graph is plotted between the work done (W) on the particle, against the speed of the particle, (v). If there are no other horizontal forces acting on the particle the graph would look like



19. A semicircular portion of radius 'r' is cut from a uniform rectangular plate as shown in figure. The distance of centre of mass 'C' of remaining plate, from point 'O' is :



- (1) $\frac{2r}{(3-\pi)}$
(2) $\frac{3r}{2(4-\pi)}$
(3) $\frac{2r}{(4+\pi)}$
(4) $\frac{2r}{3(4-\pi)}$

20. Two particles bearing mass ratio $n : 1$ are interconnected by a light inextensible string that passes over a smooth pulley. If the system is released, then the acceleration of the centre of mass of the system is :

(1) $(n - 1)^2 g$

(2) $\left(\frac{n+1}{n-1}\right)^2 g$

(3) $\left(\frac{n-1}{n+1}\right)^2 g$

(4) $\left(\frac{n+1}{n-1}\right) g$

21. A stationary body explodes into two fragments of masses m_1 and m_2 . If momentum of one fragment is p , the energy of explosion is :

(1) $\frac{p^2}{2(m_1 + m_2)}$

(2) $\frac{p^2}{2\sqrt{m_1 m_2}}$

(3) $\frac{p^2(m_1 + m_2)}{2m_1 m_2}$

(4) $\frac{p^2}{2(m_1 - m_2)}$

22. A skater of mass m standing on ice throws a stone of mass M with a velocity of v m/s in a horizontal direction. The distance over which the skater will move back (the coefficient of friction between the skater and the ice is μ) :

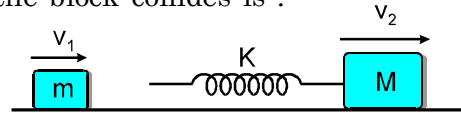
(1) $\frac{M^2 v^2}{2m\mu g}$

(2) $\frac{Mv^2}{2m^2\mu g}$

(3) $\frac{M^2 v^2}{2m^2\mu g}$

(4) $\frac{M^2 v^2}{2m^2\mu^2 g}$

23. Two blocks of masses m and M are moving with speeds v_1 and v_2 ($v_1 > v_2$) in the same direction on the frictionless surface respectively, M being ahead of m . An ideal spring of force constant k is attached to the backside of M (as shown). The maximum compression of the spring when the block collides is :



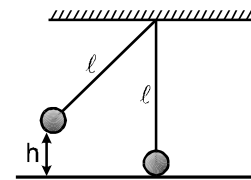
(1) $v_1 \sqrt{\frac{m}{k}}$

(2) $v_2 \sqrt{\frac{M}{k}}$

(3) $(v_1 - v_2) \sqrt{\frac{mM}{(M + m)K}}$

(4) None of above is correct.

24. In the arrangement shown, the pendulum on the left is pulled aside. It is then released and allowed to collide with other pendulum which is at rest. A perfectly inelastic collision occurs and the system rises to a height $1/4 h$. The ratio of the masses of the pendulum is :



(1) 1

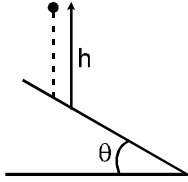
(2) 2

(3) 3

(4) 4

25. Consider the following two statements :
(a) linear momentum of the system remains constant
(b) centre of mass of the system remains at rest.
(1) a implies b and b implies a
(2) a does not imply b and b does not imply a
(3) a implies b but b does not imply a
(4) b implies a but a does not imply b
26. Consider the following two statements :
(a) Linear momentum of the system of particles is zero
(b) Kinetic energy of a system of particles is zero
(1) a implies b and b implies a
(2) a does not imply b and b does not imply a
(3) a implies b but b does not imply a
(4) b implies a but a does not imply b

27. A ball of mass m strikes the fixed inclined plane after falling through a height h . If it rebounds elastically, the impulse on the ball is :



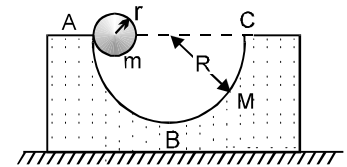
- (1) $2m \cos \theta \sqrt{2gh}$
(2) $2m \cos \theta \sqrt{gh}$
(3) $\frac{2m\sqrt{2gh}}{\cos \theta}$
(4) $2m\sqrt{2gh}$

28. A wagon filled with sand has a hole so that sand leaks through the bottom at a constant rate λ . An external force $\vec{F}$ acts on the wagon in the direction of motion. Assuming instantaneous velocity of the wagon to be $\vec{v}$ and initial mass of system to be m_0 , the force equation governing the motion of the wagon is :

- (1) $\vec{F} = m_0 \frac{d\vec{v}}{dt} + \lambda \vec{v}$
(2) $\vec{F} = m_0 \frac{d\vec{v}}{dt} - \lambda \vec{v}$
(3) $\vec{F} = (m_0 - \lambda t) \frac{d\vec{v}}{dt}$
(4) $\vec{F} = (m_0 - \lambda t) \frac{d\vec{v}}{dt} + \lambda \vec{v}$

29. A block of mass M with a semicircular track of radius R rests on a horizontal frictionless surface. A uniform cylinder of radius r and mass m is released from rest from the top point A. The cylinder slips on the semicircular frictionless track. The distance travelled by the block when the cylinder reaches the point B is :

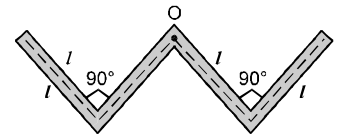
- (1) $\frac{M(R-r)}{M+m}$
(2) $\frac{m(R-r)}{M+m}$
(3) $\frac{(M+m)R}{M}$



(4) none

30. A thin rod of length $4l$, mass $4m$ is bent at the points as shown in the fig. What is the moment of inertia of the rod about the axis passing point O & perpendicular to the plane of the figure.

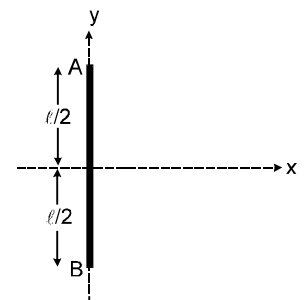
- (1) $\frac{M\ell^2}{3}$
(2) $\frac{10M\ell^2}{3}$



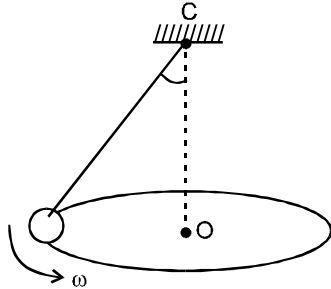
- (3) $\frac{M\ell^2}{12}$ (4) $\frac{M\ell^2}{24}$

31. A uniform rod of mass m , length ℓ is placed over a smooth horizontal surface along y-axis and is at rest as shown in figure. An impulsive force F is applied for a small time Δt along x-direction at point A. The x-coordinate of end A of the rod when the rod becomes parallel to x-axis for the first time is (initially the coordinate of centre of mass of the rod is $(0, 0)$) :-

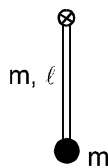
- (1) $\frac{\pi\ell}{12}$
(2) $\frac{\ell}{2} \left(1 + \frac{\pi}{12} \right)$
(3) $\frac{\ell}{2} \left(1 - \frac{\pi}{6} \right)$
(4) $\frac{\ell}{2} \left(1 + \frac{\pi}{6} \right)$



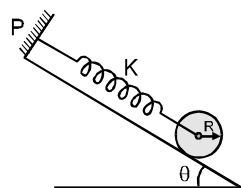
32. A conical pendulum consists of a simple pendulum moving in a horizontal circle as shown. C is the pivot, O the centre of the circle in which the pendulum bob moves and ω the constant angular velocity of the bob. If $\vec{L}$ is the angular momentum about point C, then



- (1) $\vec{L}$ is constant
(2) only direction of $\vec{L}$ is constant
(3) only magnitude of $\vec{L}$ is constant
(4) none of the above.
33. A particle is attached to the lower end of a uniform rod which is hinged at its other end as shown in the figure. The minimum speed given to the particle so that the rod performs circular motion in a vertical plane will be:



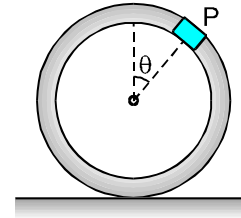
- (1) $\sqrt{5gl}$ (2) $\sqrt{4gl}$
(3) $\sqrt{4.5gl}$ (4) none of these
34. A uniform cylinder of mass M and radius R rolls without slipping down a slope of angle θ to the horizontal.



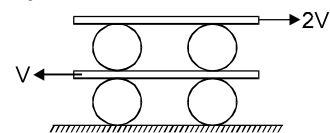
The cylinder is connected to a spring constant K while the other end of the spring is connected to a rigid support at P. The cylinder is released when the spring is unstretched. The maximum distance that the cylinder travels is

- (1) $\frac{3}{4} \frac{Mg \sin \theta}{K}$ (2) $\frac{Mg \tan \theta}{K}$
(3) $\frac{2Mg \sin \theta}{K}$ (4) $\frac{4}{3} \frac{Mg \sin \theta}{K}$

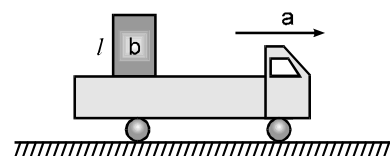
35. A small block of mass 'm' is rigidly attached at 'P' to a ring of mass '3m' and radius 'r'. The system is released from rest at $\theta = 90^\circ$ and rolls without sliding. The angular acceleration of ring just after release is –



- (1) $\frac{g}{4r}$ (2) $\frac{g}{8r}$
(3) $\frac{g}{3r}$ (4) $\frac{g}{2r}$
36. A system of uniform cylinders and plates is shown in figure. All the cylinders are identical and there is no slipping at any contact. Velocity of lower & upper plate is V and 2V respectively as shown in figure. Then the ratio of angular speed of the upper cylinders to lower cylinders is



- (1) 3 (2) 1/3
(3) 1 (4) none of these
37. A box of dimensions ℓ and b is kept on a truck moving with an acceleration a. If box does not slide, maximum acceleration for it to remain in equilibrium (w.r.t. truck) is :



- (1) $\frac{g\ell}{b}$ (2) $\frac{gb}{\ell}$
(3) g (4) none of these

38. A straight rod of length ℓ extends from $x = \alpha$ to $x = \ell + \alpha$. If the mass per unit length is $(a + bx^2)$. The gravitational force it exerts on a point mass m placed at $x = 0$ is given by

(1) $Gm \left(a \left(\frac{1}{\alpha} - \frac{1}{\alpha + \ell} \right) + b\ell \right)$

(2) $\frac{Gm(a + b\ell^2)}{\ell^2}$

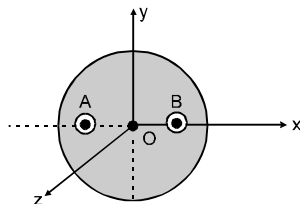
(3) $Gm \left(a \left(\frac{1}{a} - \frac{1}{a + \ell} \right) + b\ell \right)$

(4) $Gm \left(a \left(\frac{1}{\alpha + \ell} - \frac{1}{\alpha} \right) + b\ell \right)$

39. Imagine a light planet revolving around a very massive star in a circular orbit of radius R with a period of revolution T . If the gravitational force of attraction between the planet and the star is proportional to $R^{-5/2}$ then

- (1) T^2 is proportional to R^3
 (2) T^2 is proportional to $R^{7/2}$
 (3) T^2 is proportional to $R^{3/2}$
 (4) T^2 is proportional to $R^{3.75}$

40. A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii 1 unit with their centres at $A(-2, 0, 0)$ and $B(2, 0, 0)$ respectively are taken out of the solid leaving behind spherical cavities as shown in the figure.



- (1) The gravitational field due to this object at the origin is zero.
 (2) The gravitational field at the point $B(2, 0, 0)$ is zero
 (3) The gravitational field at the point $A(-2, 0, 0)$ is zero.
 (4) The gravitational field at points A , O and B is zero.

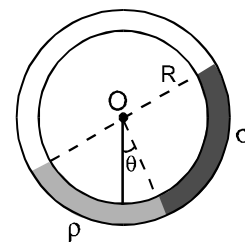
41. Following are some statements about buoyant force: (Liquid is of uniform density)

- (i) Buoyant force depends upon orientation of the concerned body inside the liquid.
 (ii) Buoyant force depends upon the density of the body immersed.
 (iii) Buoyant force depends on the fact whether the system is on moon or on the earth.
 (iv) Buoyant force depends upon the depth at which the body (fully immersed in the liquid) is placed inside the liquid.

Of these statements :

- (1) Only (i), (ii) and (iv) are correct.
 (2) Only (ii) is correct.
 (3) Only (iii) and (iv) are correct.
 (4) (i), (ii) and (iv) are incorrect.

42. A small uniform tube is bent into a circular tube of radius R and kept in the vertical plane. Equal volumes of two liquids of densities ρ and σ ($\rho > \sigma$) fill half of the tube as shown. θ is the angle which the radius passing through the interface makes with the vertical.



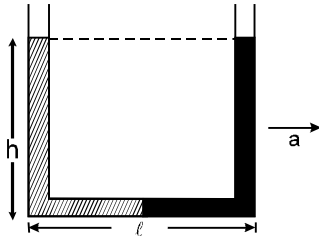
(1) $\theta = \tan^{-1} \left(\frac{\rho - \sigma}{\rho + \sigma} \right)$

(2) $\theta = \tan^{-1} \left(\frac{\sigma - \rho}{\sigma + \rho} \right)$

(3) $\theta = \tan^{-1} \left(\frac{\rho}{\rho + \sigma} \right)$

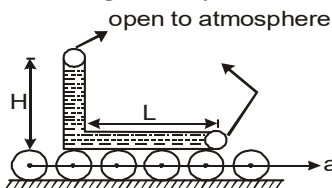
(4) $\theta = \tan^{-1} \left(\frac{\rho}{\rho - \sigma} \right)$

43. A U-tube of base length " ℓ " filled with same volume of two liquids of densities ρ and 2ρ is moving with an acceleration " a " on the horizontal plane. If the height difference between the two surfaces (open to atmosphere) becomes zero, then the height h is given by:



- (1) $\frac{a}{2g}\ell$ (2) $\frac{3a}{2g}\ell$ (3) $\frac{a}{g}\ell$ (4) $\frac{2a}{3g}\ell$

44. A narrow tube completely filled with a liquid is lying on a series of cylinders as shown in figure. Assuming no sliding between any surfaces, the value of acceleration of the cylinders for which liquid will not come out of the tube from anywhere is given by



- (1) $\frac{gH}{2L}$ (2) $\frac{gH}{L}$ (3) $\frac{2gH}{L}$ (4) $\frac{gH}{\sqrt{2}L}$

45. There is a small hole in the bottom of a fixed container containing a liquid upto height ' h '. The top of the liquid as well as the hole at the bottom are exposed to atmosphere. As the liquid comes out of the hole. (Area of the hole is ' a ' and that of the top surface is ' A ') :

(1) the top surface of the liquid accelerates with acceleration = g

(2) the top surface of the liquid accelerates

with acceleration = $g \frac{a^2}{A^2}$

(3) the top surface of the liquid retards

with retardation = $g \frac{a}{A}$

(4) the top surface of the liquid retards

with retardation = $\frac{ga^2}{A^2}$

46. Select incorrect order(s) of electronegativity of element is/are :
 (1) E.N. of F-atom on pauling scale > E.N. of N-atom on pauling scale
 (2) E.N. of Cl-atom in Cl_2O_7 > E.N. of Cl-atom in Cl_2O_5
 (3) E.N. of C-atom in CH_4 > E.N. of C-atom in CO_2
 (4) E.N. of Cu^{2+} > E.N. of Cu^+
47. I.P of Cr, Zn, Mn and Hg are 7.4eV, 6.7 eV, 10.4 eV and 9.3 eV. These I.P are not written in order then –
 (1) I.P of Zn = 10.4eV
 (2) I.P of Mn = 7.4 eV
 (3) I.P of Cr = 9.3 eV
 (4) I.P of Hg = 9.3 eV
48. In which of the following oxyacid, O-atom is not present between two central atoms?
 (1) $(\text{H}_2\text{S}_2\text{O}_8)$ (2) $\text{H}_4\text{P}_2\text{O}_7$
 (3) $(\text{H}_2\text{S}_2\text{O}_5)$ (4) $(\text{HPO}_3)_3$
49. In which of the following pairs geometry around central atom is not same ?
 (1) CH_4 , SiH_4 (2) As_4O_{10} , P_4O_{10}
 (3) CO_2 , SiO_2 (4) NO_2 , ClO_2
50. Which of the following statement is wrong:
 (1) B.P. of O-nitrophenol is greater than p – nitrophenol.
 (2) In SO_3 $\text{p}\pi - \text{d}\pi$ bond is present
 (3) In S_3O_9 no. of bridge oxygen is 3
 (4) All of these
51. Which of the following order is incorrect :
 (1) Ionic character – $\text{MCl} > \text{MCl}_2 > \text{MCl}_3$
 (2) Polarizability – $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$
 (3) Polarising power – $\text{Na}^+ < \text{K}^+ < \text{Mg}^{+2} < \text{Al}^{+3}$
 (4) Polarising power – $\text{Li}^+ < \text{Be}^{+2} < \text{B}^{+3}$
52. Which one of the following is a bronsted acid but not a bronsted base ?
 (1) H_2S (2) H_2O
 (3) HCO_3^- (4) NH_3
53. The solubility product of $\text{Mg}(\text{OH})_2$ is 1×10^{-12} . At what pH precipitation will occur in 0.01 M MgCl_2 solution?
 (1) 8 (2) 9 (3) 10 (4) 12
54. Which of the following is correct on the basis of hund's rule but not follow Pauli's principle
 (1) $\uparrow \uparrow \uparrow \uparrow$ (2) $\uparrow \downarrow \uparrow \uparrow$
 (3) $\uparrow \uparrow \uparrow \uparrow$ (4) $\uparrow \downarrow \uparrow \downarrow$
55. $aA \rightleftharpoons mB + nC$, $\Delta H = +x \text{ K.cal.}$
 If $m + n < a$ then conditions favouring forward reaction will be
 (1) Low pressure, low temperature
 (2) Low pressure, High Temperature
 (3) High pressure, low temperature
 (4) High pressure, High temperature
56. How much lime (CaO) can be obtained by heating 200 kg of limestone which is 95% pure ?
 (1) 106.4 kg (2) 53.2 kg
 (3) 10.64 kg (4) 5.32 kg
57. At high pressure, the form of Vander Wall equation becomes :
 (1) $\left(P + \frac{a}{V^2}\right)(V) = RT$
 (2) $P(V - b) = RT$
 (3) $PV = RT$
 (4) $\left(P + \frac{a}{V^2}\right)(V - b) = RT$
58. Change in entropy is positive for :
 (1) $\text{H}_2\text{O}_{(g)} \longrightarrow \text{H}_2\text{O}_{(l)}$
 (2) $\text{H}_2\text{O}_{(l)} \longrightarrow \text{H}_2\text{O}_{(s)}$
 (3) $\text{N}_{2(g)} + 3\text{H}_{2(g)} \longrightarrow 2\text{NH}_{3(g)}$
 (4) $2\text{O}_{3(g)} \longrightarrow 3\text{O}_{2(g)}$
59. In acidic medium moles of KMnO_4 required to oxidise 1.5 moles of Cu_2S will be :
 (1) 1.2 mol (2) 1.6 mol
 (3) 2.1 mol (4) 2.4 mol

60. If D_T and D_0 are the theoretical and observed vapour densities at a definite temperature and α be the degree of dissociation of a substance. Then, α in the terms of D_0 , D_T and n (number of moles of product formed from 1 mole reactant) is calculated by the formula :-

$$(1) \alpha = \frac{D_0 - D_T}{(1-n)D_T} \quad (2) \alpha = \frac{D_T - D_0}{(n-1)D_T}$$

$$(3) \alpha = \frac{D_T - D_0}{(n-1)D_0} \quad (4) \alpha = \frac{D_0 - D_T}{(n-1)D_T}$$

61. Which of the following is least likely to behave as Lewis Base ?

(1) H_2O (2) NH_3 (3) BF_3 (4) OH^-

62. From the following the largest number of atoms are in

(1) 36g H_2O (2) 28g CO
(3) 46g C_2H_5OH (4) 54g N_2O_5

63. Which of the following electronic configurations is not possible according to Hund's rule

(1) $1s^2 2s^2$
(2) $1s^2 2s^1$
(3) $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$
(4) $1s^2 2s^2 2p_x^2$

64. The ratio of the radii of the first three Bohr orbits is -

(1) $1 : \frac{1}{2} : \frac{1}{3}$ (2) $1 : 2 : 3$
(3) $1 : 4 : 9$ (4) $1 : 8 : 27$

65. The units of constant 'a' in Vander Waal's equation is :

(1) $dm^6 atm mol^{-2}$
(2) $dm^3 atm mol^{-1}$
(3) $dm atm mol^{-1}$
(4) $atm mol^{-1}$

66. The volumes of two HCl solutions A(0.5 M) and B(0.1 M) to be mixed for preparing 2L of 0.2 M HCl are :-

(1) 0.5 L of A + 1.5 L of B
(2) 1.5 L of A + 0.5 L of B
(3) 1 L of A + 1 L of B
(4) 0.75 L of A + 1.25 L of B

67. Incorrect statement is :

(1) Bond form by s-s overlapping is always weakest
(2) SiO_2 is 3-D network solid
(3) Coaxial overlapping always produce bond
(4) Hybrid orbitals produce bond

68. Which of the following has been arranged in order of decreasing bond length :

(1) $P-O > Cl-O > S-O$
(2) $P-O > S-O > Cl-O$
(3) $S-O > Cl-O > P-O$
(4) $Cl-O > S-O > P-O$

69. The correct sequence of decreases in the bond angles of the following hydrides is :

(1) $NH_3 > PH_3 > AsH_3 > SbH_3$
(2) $NH_3 > AsH_3 > PH_3 > SbH_3$
(3) $SbH_3 > AsH_3 > PH_3 > NH_3$
(4) $PH_3 > NH_3 > AsH_3 > SbH_3$

70. Which of the following species is paramagnetic?

(1) NO^- (2) O_2^{2-} (3) CN^- (4) CO

71. From the given set of species, point out the species from each set having least atomic radius :

(A) F^- , Na^+ , Mg^{+2} (B) Ni , Cu , Zn
(C) N^{-3} , Cs^+ , H^- (D) Li , He , Be^{+2}
(1) Mg^{+2} , Ni , N^{-3} , Be^{+2}
(2) Na^+ , Cu , Cr^+ , Li
(3) F^- , Cu , N^{-3} , He
(4) Na^+ , Ni , H^- , He

72. Incorrect order of ionic radius is :

(1) $Na^+ > Mg^{+2} > Al^{+3}$
(2) $V^{+2} > V^{+3} > V^{+4} > V^{+5}$
(3) $K^+ > Sc^{+3} > V^{+5} > Mn^{+7}$
(4) $O^{-2} > Rb^+ > Sr^{+2} > Cs^+$

73. When 2 L of CO_2 is heated with graphite, volume of gases collected is 3 L, calculate number of moles of CO produced at STP?

- (1) $\frac{1}{22.4}$ (2) $\frac{2}{22.4}$
(3) $\frac{3}{22.4}$ (4) $\frac{4}{22.4}$

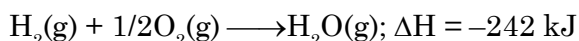
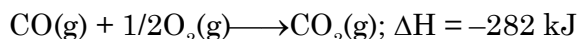
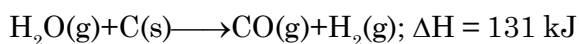
74. 3.00 mol of PCl_5 kept in 1L closed vessel was allowed to attain equilibrium at 380 K. If $K_C = 1.80$ then calculate equilibrium concentrations of PCl_3 & Cl_2 respectively :

- (1) 1.59 M, 1.59 M
(2) 2.28 M, 2.28 M
(3) 3 M, 3 M
(4) 2.28 M, 3 M

75. The solubility of AgCl is minimum in :

- (1) AgNO_3 (0.1M) (2) H_2O (ℓ)
(3) NaCl (0.4M) (4) BaCl_2 (0.3M)

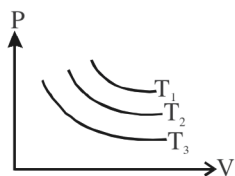
76. Based on the following thermochemical equations



The value of X will be :

- (1) -393 kJ (2) -655 kJ
(3) +393 kJ (4) +655 kJ

77. Following P vs V curve is plotted :



What is the relation between T_1 , T_2 and T_3 ?

- (1) $T_1 = T_2 = T_3$
(2) $T_1 > T_2 > T_3$
(3) $T_1 < T_2 < T_3$
(4) None of the above

78. At identical temperature and pressure, the rate of diffusion of hydrogen gas is $3\sqrt{3}$ times that of a gaseous hydrocarbon having molecular formula $\text{C}_n\text{H}_{2n-2}$. What is the value of n?

- (1) 1 (2) 4 (3) 3 (4) 8

79. In which of the following transition photon of maximum energy is absorbed :

- (1) 1 to 2 (2) 2 to 3
(3) 1 to ∞ (4) ∞ to 1

80. If electron moves from an excited state to ground state in multistep in hydrogen spectrum lines in visible region are 4. Then number of lines in infra red region is :

- (1) 4 (2) 6 (3) 5 (4) 7

81. For the configuration $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$ which rule is violated ?

- (1) Aufbau principle
(2) $(n + \ell)$ Rule
(3) Pauli 's Rule
(4) Hund's Rule

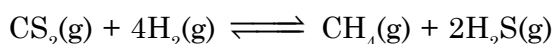
82. Numbers of gram atoms of oxygen present in 0.8mol of $\text{C}_4\text{H}_6\text{O}_6$ would be :

- (1) 3.6 (2) 1.8 (3) 4.8 (4) 2.4

83. A gas is found to contain 2.8 g nitrogen and 3.2g oxygen, simplest formula of the compound is :

- (1) N_2O (2) NO
(3) N_2O_3 (4) NO_2

84. What is the unit of K_p for the reaction ?

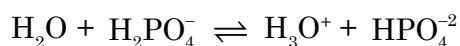


- (1) atm (2) atm^{+2}
(3) atm^{-2} (4) atm^{-1}

85. The oxidation state of molybdenum in its oxo complex species $[\text{Mo}_2\text{O}_4(\text{C}_2\text{H}_4)_2(\text{H}_2\text{O}_2)]^{2-}$ is

- (1) +2
(2) +4
(3) +3
(4) +5

86. $\text{HF} + \text{HClO}_4 \rightleftharpoons \text{H}_2\text{F}^+ + \text{ClO}_4^-$ and



Select out the group of acidic species lying on the right side of the equilibrium :



87. 0.1M CH_3COOH is titrated against 0.1M NaOH . What would be the difference in pH between 1/4 and 3/4 stages of neutralisation of acid :-

(1) $2\log \frac{3}{4}$ (2) $2\log \frac{1}{4}$

(3) $\log \frac{8}{3}$ (4) $2\log 3$

88. For the reaction $\text{FeCO}_{3(s)} \longrightarrow \text{FeO}_{(s)} + \text{CO}_{2(g)}$
 $\Delta H = 82.8 \text{ kJ}$ at 25°C . What is ΔU at 25°C ?

(1) 82.8 kJ (2) 80.32 kJ

(3) -2394.77 kJ (4) 85.28 kJ

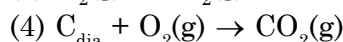
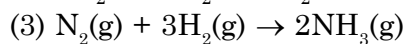
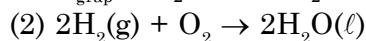
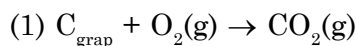
89. Which of the following reaction satisfy following conditions?

(a) Heat of combustion reaction

(b) Heat of formation reaction

(c) An exothermic reaction

(d) Not neutralization reaction



90. What will be ΔS for reaction : $2\text{A} + 3\text{B} \rightarrow 4\text{C} + 5\text{D}$

$$\Delta S_A^\circ = 100 \frac{\text{J}}{\text{mol} \times \text{K}}$$

$$\Delta S_C^\circ = 200 \frac{\text{J}}{\text{mol} \times \text{K}}$$

$$\Delta S_B^\circ = 120 \frac{\text{J}}{\text{mol} \times \text{K}}$$

$$\Delta S_D^\circ = 150 \frac{\text{J}}{\text{mol} \times \text{K}}$$

(1) $105 \frac{\text{J}}{\text{K}}$

(2) $990 \frac{\text{J}}{\text{K}}$

(3) $300 \frac{\text{J}}{\text{K}}$

(4) $130 \frac{\text{J}}{\text{K}}$

91. Which of the following is not applicable to living organisms?
- Self replicating and evolving
 - Self replicating and capable of responding to external stimuli
 - Self replicating and non-regulative interactive system
 - Metabolism
92. Incorrect about growth :-
- Growth and reproduction are mutually exclusive events in higher plants and animals
 - In living organisms growth occur from inside
 - Non-living substances may also show growth but from outside
 - None of the above
93. What is correct regarding binomial nomenclature?
- First name is specific epithet
 - Second name is of class
 - Developed by John Ray
 - None of these
94. Which information is not conveyed on label of a herbarium sheet?
- Collector name
 - Date and place of collection
 - Botanical name along with local / English name
 - Details of other related members
95. Which of the following statement is false?
- No non-living object exhibits metabolism
 - No non-living object shows reproduction
 - No non-living object shows growth
 - None is false, all are true statements
96. Identify the correct option regarding *Mangifera indica* (mango) :

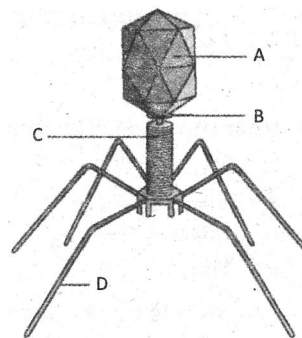
Family	Order	Class
(1) Anacardiaceae	Sapindales	Dicotyledonae
(2) Sapindales	Anacardiaceae	Dicotyledonae
(3) Anacardiaceae	Sapindales	Monocotyledonae
(4) Sapindales	Anacardiaceae	Monocotyledonae

97. Fill in the blanks :-

Organism	Class	Order
Man	A	Primata
Housefly	Insecta	B

A and Bare :-

- Chordata and Muscidae
 - Mammalia and Muscidae
 - Mammalia and Diptera
 - Vertebrata and Diptera
98. Who was the first to classify plants into trees, shrubs and herbs?
- Aristotle
 - R.H. Whittaker
 - Carolus Linnaeus
 - Pliny
99. In a bacteriophage A-D are labelled. Select the correct information.



- A - Head, B - Tail fibres
 - A - Head, D - Sheath
 - B - Collar, D - Tail fibres
 - C - Sheath, A - Collar
100. Identify the correct match :-
- Nostoc—Photosynthetic autotroph
 - Euglena—Marine protist
 - Ustilago—Rust fungi
 - Gonyaulax—saprophytic protist

101. Identify the mismatched one :-

(1) Desmids	These are members of kingdom Protista
(2) Slime moulds	Under suitable conditions these form Plasmodium
(3) Ciliated protozoans	These possess two cilia-one large and another small
(4) Mycoplasma	These can survive without oxygen

102. Red tide is the members of :-

- (1) Chrysophyta
- (2) Euglenoids
- (3) Dinoflagellata
- (4) Slime moulds

103. Dispersal of spores of slime moulds occur through :-

- (1) Air current
- (2) Water
- (3) Insects
- (4) Crustaceans

104. Identify the incorrect match :-

Class of Fungi	Sexual / Asexual spores	Example
(1) Basidiomycetes	Oospores	Rhizopus
(2) Ascomycetes	Ascospores	Neurospora
(3) Basidiomycetes	Conidia	Agaricus
(4) Deuteromycetes	Conidia	Alternaria

105. Stored material in the class of algae to which Fucus and Dictyota belong, is :-

- (1) Floridean starch
- (2) Starch
- (3) Mannitol and Laminarin
- (4) Algin

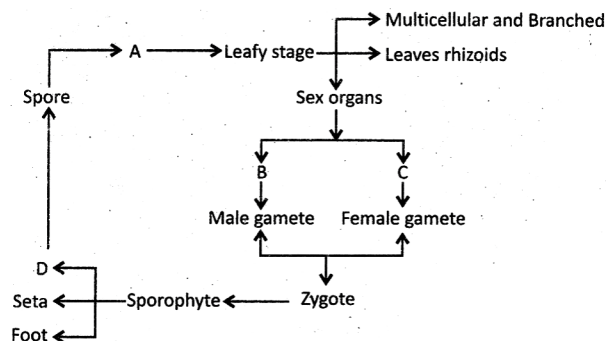
106. How many of the given pteridophytes are heterosporous-Equisetum, Pteris, Adiantum, Lycopodium, Selaginella, Dryopteris?

- (1) 1
- (2) 2
- (3) 3
- (4) 4

107. Gemma the green, multicellular, asexual buds develop in :

- (1) Marchantia
- (2) Funaria
- (3) Selaginella
- (4) Polytrichum

108. Life cycle of a moss is given :-



Select the wrong statement :-

- (1) A is protonema, it represents gametophyte stage
- (2) B is antheridia and is male sex organ -
- (3) C is carpel and is female sex organ
- (4) D is capsule which produces spores within

109. Pollen grains in gymnosperms and antherozoids in pteridophytes respectively spread through :-

- (1) Water, Water
- (2) Wind, Water
- (3) Animals, Wind
- (4) Wind, Animals

110. Double fertilization in angiosperms results in the formation of :-

- (1) Diploid zygote and triploid endosperm nucleus
- (2) Diploid zygote and haploid endosperm nucleus
- (3) Diploid zygote only.
- (4) Triploid primary endosperm nucleus only

111. Which aestivation is found in calyx of members of Fabaceae?

- (1) Valvate only
- (2) Quincuncial only
- (3) Valvate or quincuncial
- (4) Valvate or imbricate

112. Adventitious roots are found in all of the following except :

- (1) Grass (2) Banyan
(3) Monstera (4) Mustard

113. Out of the following plants, how many possess stilt roots?

[Banyan, Neem, Carrot, Maize, Mustard, Monstera, Sugarcane]

- (1) 2 (2) 3 (3) 4 (4) 5

114. Midrib is observed in which part of leaf :

- (1) Leaf base (2) Lamina
(3) Petiole (4) Stipule

115. Which of the following is not the part of leaf?

- (1) Lamina (2) Petiole
(3) Pedicel (4) Leaf base

116. In which of the following plants, tendril is modification of leaf?

- (1) Pea (2) Grapevine
(3) Pumpkin (4) Cucumber

117. When stamens are attached to perianth, such condition is known as :-

- (1) Epipetalous (2) Epiphyllous
(3) Polyandrous (4) Diadelphous

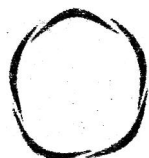
118. Which set of plants are having flowers with half inferior ovary?

- (1) Plum, Peach
(2) Cucumber, Mustard
(3) Brinjal, Mustard
(4) Plum, Brinjal

119. Ovary is enclosed by thalamus completely in :

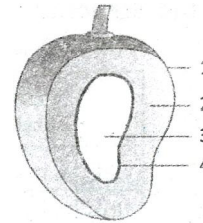
- (1) China rose (2) Rose
(3) Guava (4) Plum

120. Identify the aestivation given in diagram with suitable example :



- (1) Imbricate, Bean
(2) Imbricate, Cassia
(3) Vexillary, Gulmohar
(4) Vexillary, Pea

121. 1-4 are indicated in section of mango. Select the option with correct identification :



(1) 1 = Testa, 2 = Tegmen, 3 = Endosperm, 4 = Seed

(2) 1 = Testa, 2 = Tegmen, 3 = Seed, 4 = Endosperm

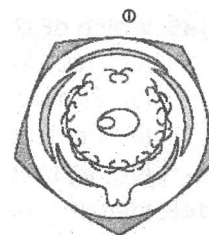
(3) 1 = Epicarp, 2 = Mesocarp, 3 = Seed, 4 = Endocarp

(4) 1 = Pericarp, 2 = Mesocarp, 3 = Seed, 4 = Endocarp

122. Perianth with 6 tepals (3+3), arranged in valvate aestivation is seen in members of angiospermic family :

- (1) Solanaceae (2) Liliaceae
(3) Fabaceae (4) Ranunculaceae

123. On the basis of given floral diagram, identify the incorrect statement :



(1) 10 stamens are arranged in two groups

(2) Carpel is single

(3) Polysepalous with imbricate aestivation

(4) Polypetalous with vexillary aestivation

124. How many of the following are examples of lateral meristem?

[Cork cambium, intercalary meristem, Intrafascicular cambium, interfascicular cambium, Apical meristem]

- (1) 4 (2) 3 (3) 2 (4) 1

- 125.** Cell wall of sclerenchymatous fibres and sclereids are respectively thickened by:
- (1) Cellulose, cellulose
 - (2) Cellulose, hemicelluloses
 - (3) Lignin, cellulose
 - (4) Lignin, lignin
- 126.** Nucleus of cell A controls the activities of cell B in secondary phloem. A and B are respectively :
- (1) Phloem parenchyma, phloem fibres
 - (2) Bast fibres, companion cells
 - (3) Companion cell, sieve tube
 - (4) Sieve tube, companion cell
- 127.** Fibres for commercial use, obtained from which plant are not bast fibres :
- (1) Jute
 - (2) Cotton
 - (3) Hemp
 - (4) Flax
- 128.** Root hairs and trichomes are respectively :
- (1) Unicellular and unicellular
 - (2) Unicellular and multicellular
 - (3) Multicellular and unicellular
 - (4) Multicellular and multicellular
- 129.** In grasses, shape of guard cells in stomata is :
- (1) Dumb—bell shaped
 - (2) Bean shaped
 - (3) Spindle shaped
 - (4) Pyriform shaped
- 130.** Typical dicot stem possess open vascular bundles. What is incorrect about such vascular bundles?
- (1) Inter-fascicular cambium is present between xylem and phloem
 - (2) It may show secondary growth
 - (3) It may give rise to secondary xylem and secondary phloem both
 - (4) Cambium of such bundles is example of secondary meristem
- 131.** Cambium ring in dicot root is initially:
- (1) Wavy
 - (2) Circular
 - (3) Discontinuous
 - (4) Crescentric
- 132.** During secondary growth in dicot stem, cork cambium has cut off few cells which are impermeable to water. These cells have been cut off towards :
- (1) Outer side
 - (2) Inner side
 - (3) Lateral side
 - (4) In all directions
- 133.** During secondary growth in root :
- (1) Girth of root is not increased
 - (2) Cambium is already present but less differentiated
 - (3) Cambium is already present and is highly differentiated
 - (4) Cambium appears only at the time of secondary growth
- 134.** Incorrect about sapwood :
- (1) Outer to heartwood
 - (2) Part of annual ring
 - (3) Functionally active
 - (4) Performs conduction of water and minerals
- 135.** Which of the following is not a correct statement?
- (1) Algal cells possess 80 S ribosomes
 - (2) Slime layer is outer most layer of cell envelope in a bacterial cell
 - (3) Mesosomes are found in protistan cells
 - (4) Lipids are more abundant in a plasma membrane than proteins
- 136.** Select the wrong statement :
- (1) Cell theory was modified by Rudolf Virchow in 1839
 - (2) PPLO is a prokaryote
 - (3) Genophore or incipient nucleus is found in prokaryotic cells
 - (4) First live cell was discovered by A.V. Leeuwenhoek
- 137.** How many of the given cell organelles are found only in prokaryotic cells?
- [Vacuoles, Mesosomes, Ribosomes, Flagella, Cell wall, Mitochondria]
- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4

138. A cell organelle in prokaryotes is concerned with storage of reserve food materials e.g., cyanophycean granules, etc. It is a non-membrane bound cell organelle, it is :

- (1) Gas vacuole (2) Mesosome
(3) Plasmid (4) Inclusion body

139. Part in common sharing of two plant cells, at which cell wall is absent :

- (1) Vacuole
(2) Plasmodesmata
(3) Polysomes
(4) Middle lamella

140. Main constituents of middle lamella are:

- (1) Cellulose
(2) Hemicellulose
(3) Galactans
(4) Calcium and magnesium pectate

141. pH inside lysosome is:

- (1) Acidic, 5 (2) Acidic, 6.8
(3) Alkaline, 7.4 (4) Alkaline, 8.8

142. In Chlamydomonas, chloroplast is A..... shaped and the number isB.....

- (1) A = cup, B = few
(2) A = girdle, B = few
(3) A = cup, B = 1
(4) A = girdle, B = many

143. In eukaryotic flagella, membrane bound central part is termed as:

- (1) Shaft (2) Central sheath
(3) Axoneme (4) Spoke

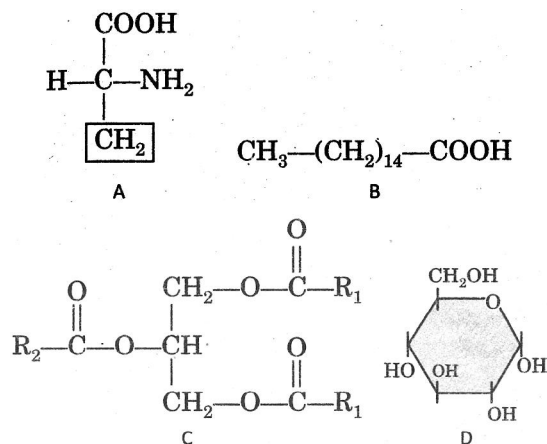
144. Which of the following is incorrectly matched between column I and Column II?

Column I : (Organism)	Column II : (Shape of chloroplast)
(1) Spirogyra	Ribbon shaped
(2) Chlamydomonas	Cup shaped
(3) Oedogonium	Reticular
(4) Ulothrix	Spindle shaped

145. Which of the following cells lack nucleus?

- (1) Mature-RBC (2) Amoeba
(3) Sieve tube (4) Both (1) and (3)

146. Identify the correct information about the given diagrams?



- (1) B is triglyceride and is a type of lipid
(2) D is fructose and is a carbohydrate
(3) A is an amino acid and participates in protein formation
(4) C is a cholesterol and it is a type of lipid

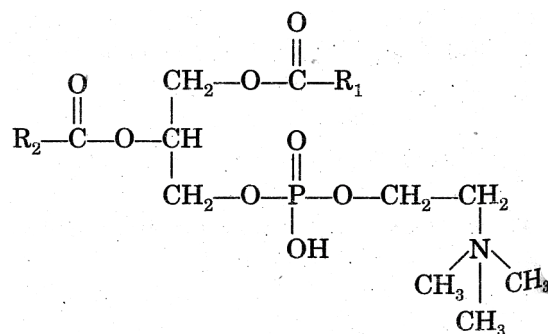
147. Which of the following polysaccharide is found in exoskeleton of arthropods and cell wall of fungi?

- (1) Glycogen (2) Chitin
(3) Cellulose (4) Inulin

148. Which of the following is having low molecular weight but found in acid insoluble fraction?

- (1) Proteins (2) Carbohydrates
(3) Nucleic acids (4) Lipids

149. Identify the given compound :

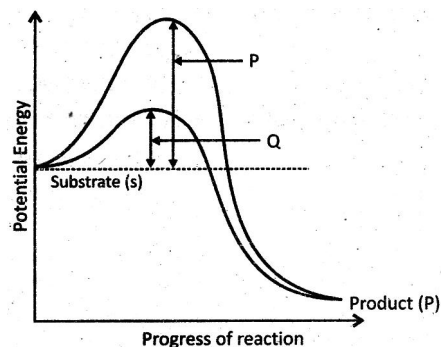


- (1) Lecithin (2) Cellulose
(3) Fatty acid (4) Triglyceride

150. Incorrect about gingely oil is :

- (1) It possess fatty acids
- (2) It lacks glycerol
- (3) It is having low melting point
- (4) It is a complex fatty acid

151. A graph between potential energy of substrate and progress of reaction is drawn, P and Q are two peaks of activation energy. Select the correct statement :

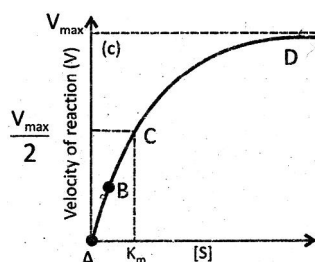


- (1) P is activation energy with enzyme
- (2) P is activation energy with holoenzyme
- (3) Q is activation energy with holoenzyme
- (4) Q is activation energy with apoenzyme

152. Which mineral act as cofactor for carbonic anhydrase enzyme?

- (1) Cu
- (2) Fe
- (3) Zn
- (4) Pb

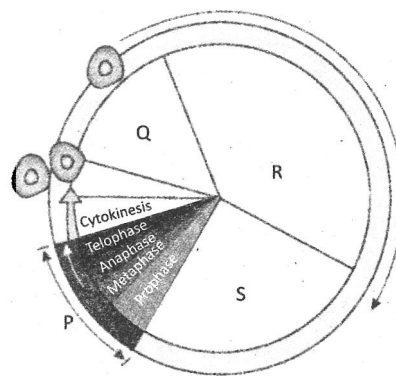
153. A, B, C and D are four points on the curve drawn between velocity of reaction (V) and concentration of substance (S) on enzyme activity. Select the true statement :



- (1) At point D, enzyme activity is optimum
- (2) At point A, enzyme activity is half of optimum
- (3) At point C, all substrate molecules are engaged
- (4) At point B, most of the enzyme molecules are engaged

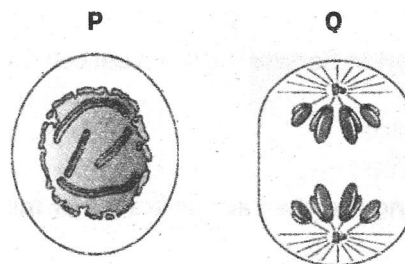
154. Diagrammatic view of cell cycle is given.

Select the correct option :-



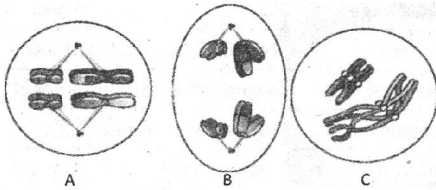
- (1) P is representing M phase, in which the cell prepares for cell division
- (2) Q is representing G_1 phase in which cell synthesise DNA
- (3) R is representing G_1 phase in which the cell prepares itself for DNA replication
- (4) S is representing G_2 phase in which the cell prepares itself for cell division

155. P and Q are two stages of mitosis. Correctly identify them:



	P	Q
(1)	Prophase	Metaphase
(2)	Metaphase	Anaphase
(3)	Telophase	Metaphase
(4)	Prophase	Anaphase

156. A, B and C are three stages in the meiosis-I as given in diagrams.



- (1) $A \rightarrow C \rightarrow B$ (2) $A \rightarrow B \rightarrow C$
(3) $C \rightarrow A \rightarrow B$ (4) $C \rightarrow B \rightarrow A$

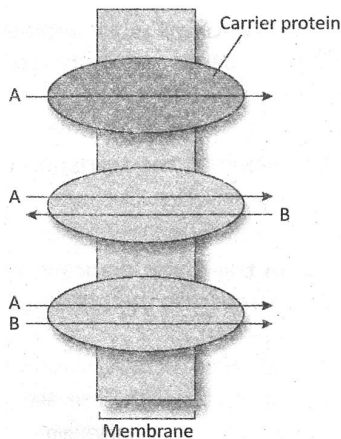
157. In plant cells, cytokinesis occur by a method in which....A.....develops from.....B.....toC.....

	A	B	C
(1)	Cell furrow	Centre	Periphery
(2)	Cell furrow	Periphery	Centre
(3)	Cell plate	Centre	Periphery
(4)	Cell plate	Periphery	Centre

158. In xylem of rooted plants, transport of water and minerals is..... and respectively.

- (1) Unidirectional, Unidirectional
(2) Unidirectional, Multidirectional
(3) Multidirectional, Unidirectional
(4) Multidirectional, Multidirectional

159. On the basis of direction of transport of substances, facilitated transport is of three types. Select the option with their correct identification



	Type-I	Type-II	Type-III
(1)	Antiport	Uniport	Symport
(2)	Antiport	Symport	Uniport
(3)	Uniport	Antiport	Symport
(4)	Uniport	Symport	Antiport

160. Storage organs receive organic nutrients from :

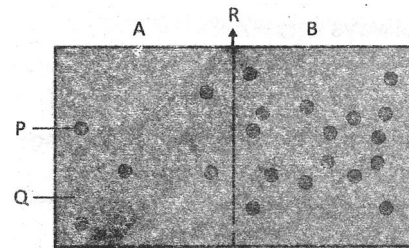
- (1) Roots (2) Leaves
(3) Buds (4) Flower

161. How many of the following statements are correct regarding facilitated transport?

- (a) Membrane proteins required.
(b) It can work against concentration gradient
(c) ATPs are required.
(d) Lipids soluble substances are commonly transported.

- (1) One statement
(2) Two statements
(3) Three statements
(4) All are correct

162. Two solutions of different concentrations (A and B) are separated by means of a membrane (R). Which of the following option is correct (take P = Solute and Q = Water)?



	Nature of membrane (R)	Net movement of	Direction of movement
(1)	Permeable	P	$A \rightarrow B$
(2)	Permeable	Q	$A \rightarrow B$
(3)	Semi-permeable	Q	$B \rightarrow A$
(4)	Semi-permeable	Q	$A \rightarrow B$

163. Typically the water content of most of the herbaceous plants is :

- (1) 65-70% (2) 75-80%
(3) 85-90% (4) 95-100%

164. Transpiration pull is developed in vessels because water possess :

- (1) High tensile strength and high capillarity
(2) High tensile strength and low capillarity
(3) Low tensile strength and high capillarity
(4) Low tensile strength and low capillarity

165. Girdling (or ringing) experiments prove that :

- (1) Xylem tissues are located inner to phloem
- (2) Pith is located inner to xylem
- (3) Inorganic minerals are translocated via xylem elements
- (4) Organic content is mainly translocated through phloem and that is usually unidirectional

166. Match the columns :

Column-A		Column-B	
A.	Hydroponics	1.	Zinc
B.	Trace element	2.	Fe
C.	Activation of catalase	3.	Julius von Sachs
D.	Dark Reaction	4.	Calvin

- (1) A-3, B-2, C-1, D-4
- (2) A-4, B-3, C-2, D-1
- (3) A-3, B-1, C-2, D-4
- (4) A-4, B-2, C-3, D-1

167. Match the column-I with Column-II :

Column-I		Column-II	
A.	Micronutrients	1.	Fe, Mn
B.	Macronutrients	2.	Nitrosomonas
C.	Nitrification	3.	K, S
D.	Denitrification	4.	Thiobacillus

- (1) 1-C, 2-A, 3-D, 4-B
- (2) 1-C, 2-A, 3-B, 4-D
- (3) 1-A, 2-C, 3-D, 4-B
- (4) 1-A, 2-C, 3-B, 4-D

168. Match the items in column-I with those in column-II :

Column-I		Column-II	
A.	Ca	1.	Chlorosis
B.	Mo	2.	Delayed flowering
C.	Fe	3.	Necrosis
D.	Cl	4.	Photolysis of water

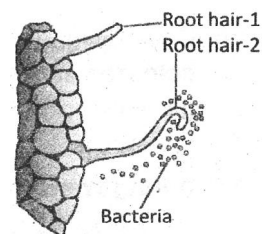
- (1) A-3, B-2, C-1, D-4
- (2) A-2, B-3, C-4, D-1
- (3) A-3, B-2, C-4, D-1
- (4) A-1, B-4, C-3, D-2

169. Match the following :

Column-A		Column-B	
A.	Photolysis of water	1.	Zn
B.	Component of nitrogenase	2.	Mn
C.	Component of ferredoxin	3.	S
D.	Biosynthesis of auxin	4.	Mo

- (1) A-2, B-4, C-1, D-5, E-3
- (2) A-2, B-4, C-3, D-1, E-5
- (3) A-3, B-1, C-2, D-4, E-5
- (4) A-1, B-3, C-2, D-5, E-4

170. Which of the following is correct?



- (1) Root hair 1 is infected but root hair 2 is not
- (2) Only root hair 2 is infected as bacteria are around it
- (3) Only root hair 2 is infected as it is curled
- (4) Both Root hair 1 and 2 are infected

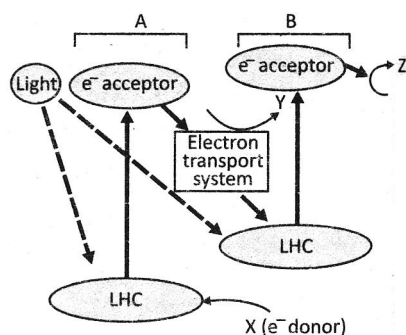
171. Various steps of mechanism of nodule formation in legumes are given below. Arrange them in correct order. A - Formation of bacteroids B - Rhizobium multiply around root hairs C - Infection thread release bacteria into the cells of inner cortex D - Curling of root hair E - Formation of nodules F - Establishment of direct vascular connection with host

- (1) B → D → C → E → A → F
- (2) D → B → E → A → F → C
- (3) D → B → A → C → F → E
- (4) B → D → C → A → E → F

172. A mint plant, a mouse and a burning candle were placed in a bell jar. Which of the following is not expected?

- (1) Mouse stayed alive and candle continued to burn
- (2) Plant will die but the mouse will stay alive
- (3) Plant and mouse stay alive
- (4) Candle continue to burn and plant will stay alive

173. Non-cyclic photo-phosphorylation is shown in the given diagram, select the option with correct identification.



	A	B	X	Y	Z
(1)	Photosystem-I	Photosystem-II	Water	ATP	NADPH
(2)	Photosystem-I	Photosystem-II	NADPH	ATP	Water
(3)	Photosystem-II	Photosystem-I	Water	NADPH	ATP
(4)	Photosystem-II	Photosystem-I	Water	ATP	NADPH

174. Select the unmatched pair?

- (1) Oxaloacetic acid-4 carbon acid
- (2) RuBisCO-Protein
- (3) Tomato- C_4 plant
- (4) Maize-Kranz anatomy

175. Match the items in column I with those in column II :

Column-I	Column-II
1. RuBP carboxylase	A. CO_2 fixation in C_4 plants
2. PEP case	B. Thylakoid membrane
3. ATP ase	C. Most abundant enzyme in world

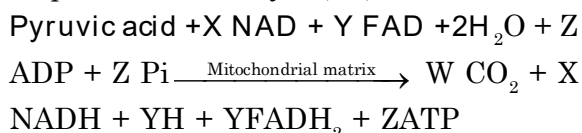
- (1) 1-C, 2-B, 3-A
- (2) 1-C, 2-A, 3-B
- (3) 1-A, 2-B, 3-C
- (4) 1-A, 2-C, 3-B

176. Match the columns :

Column-A	Column-B
1. Glycolysis	A. Cytoplasm in eukaryotes
2. ETS	B. Mitochondrial matrix
3. Krebs Cycle	C. Inner Mitochondrial membrane
	D. Outer Mitochondrial membrane
	E. Cytoplasm in prokaryotes

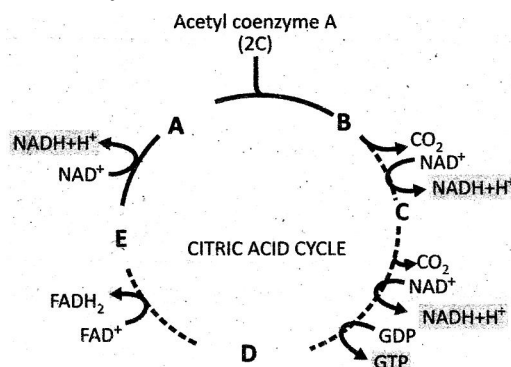
- (1) 1-A, E, 2-D, 3-B
- (2) 1-A, 2-C, 3-B
- (3) 1-A, E, 2-C, 3-B
- (4) 1-E, 2-D, 3-C

177. Following reaction indicates the complete breakdown of pyruvic acid in aerobic respiration. Identify W, X, Y and Z.



	W	X	Y	Z
(1)	3	3	1	0
(2)	3	4	1	1
(3)	6	3	2	0
(4)	6	4	1	1

178. Citric acid cycle is shown in the figure. Identify A-E.

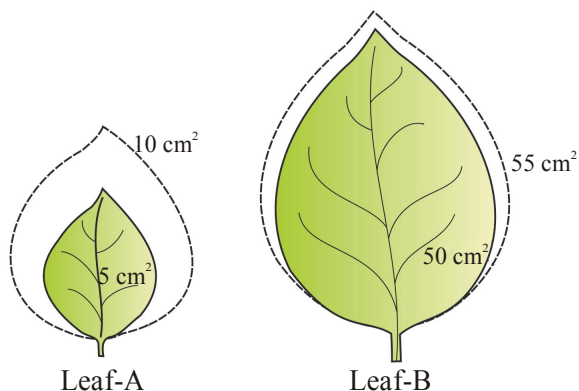


	A	B	C	D	E
(1)	Citric Acid	Oxaloacetic acid	Succinic acid	Pyruvic acid	Malic acid
(2)	Oxaloacetic acid	Citric acid	Succinic acid	Malic acid	Succinyl CoA
(3)	Oxaloacetic acid	Citric acid	α -keto-glutaric	Succinic acid	Malic acid
(4)	Oxaloacetic acid	Citric acid	α -keto-glutaric	Pyruvic acid	Malic acid

179. Apical dominance can be overcome by application of :-

- (1) GA (2) Ethylene
(3) Cytokinin (4) IAA

180. The figure given below show growth of two leaves in one week. Select the correct option, if RGR = relative growth rate and AGR = absolute growth rate.



Options	RGR of A	AGR of A	RGR of B	AGR of B
(1)	5%	100	5%	100
(2)	50%	5	50%	5
(3)	10%	5	100%	5
(4)	100%	5	100%	5